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Detection algorithm for magnetic dipole target based on CEEMDAN and pattern recognition

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Abstract

Due to the physical characteristics that the magnetic dipole target signal (MDTS) decays with the third power of distance and the fact that the measured data contain usually environmental magnetic noise such as diurnal variation noise and cultural noise, it is very difficult to detect long-distance magnetic targets. In this paper, the complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) algorithm which is an improved version of empirical mode decomposition (EMD) algorithm and pattern recognition algorithm are combined to construct a novel magnetic target detection algorithm(CEEMDAN-PR). CEEMDAN algorithm can effectively decompose the measured magnetic signal into multiple intrinsic mode functions (IMFs), and reduce the aliasing effect between modes. Then, the pattern formed by the characteristics of magnetic dipole signal is used to match the signal reconstructed by the sum of several IMFs to obtain the optimal reconstructed signal. Some simulation tests illustrate that this algorithm has good detection performance.

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Keywords: Magnetic dipole signal detection; EMD;CEEMDAN; pattern recognition.

1. Introduction

When the distance between the magnetic target and the sensor is more than three times of the target size, the magnetic target can be regarded as a magnetic dipole, and the magnetic field generated by it can be approximately expressed by the magnetic dipole field formula. This kind of detection technique utilizing the characteristic of magnetic dipole is often used in airborne magnetic survey, detection of UXO, prospecting, classification of land

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vehicles and so on[1]. In view of its wide applications, the detection method has great research value. In the process of detection, the target signal collected by magnetic sensor is usually affected by the geomagnetic noise and man-made magnetic noise, which aggravates the difficulty of magnetic dipole target signal detection. Therefore, the research of detection methods generally starts from two aspects: signal extraction and noise reduction.

In the aspect of signal extraction, the orthonormal basis function (OBF) method[2] is proposed to find a MDTS from a measured sequence, which can extract the pseudo signal energy when the signal waveform itself is not known. A target is judged to be present when the signal energy exceeds a prescribed threshold value. In addition to the classical OBF method, there are many alternative approaches, including the minimum entropy detector[3], high-order crossing approach[4], stochastic resonance method[5], signal space method[6], modified orthonormal basis function method[7], and so on. Besides, many filtering schemes are incorporated into the detection methods to reduce the effect of magnetic noise, such as the whitening filtering[8], and adaptive coherent noise suppression[9].

In addition to the detection of a target signal, if the signal waveform can be restored to the maximum extent, it will be conducive to further target recognition. EMD algorithm is a method for processing non-stationary and nonlinear signals proposed by Huang *et al*. Because of the mode aliasing problem, EMD algorithm is improved to CEEMDAN algorithm. In this paper, CEEMDAN algorithm will be used to design a magnetic dipole target detection algorithm. At the same time, considering that the selection of modes that restricts the quality of reconstructed signal is a difficult point for EMD series algorithm, this paper introduces a pattern recognition algorithm obtained from the magnetic dipole formula to determine the reconstructed signal consisting of several IMFs. In this paper, the detection algorithm based on CEEMDAN and pattern recognition is called CEEMDAN-PR algorithm. In order to facilitate the comparison, EMD and pattern recognition are combined to form EMD-PR algorithm and is used in the simulation experiments to prove the robustness and high detection performance of the proposed algorithm.

2. Detection algorithm

The detection algorithm consists of three parts: 1) the CEEMDAN algorithm decomposes the measured signal into several intrinsic mode functions; 2) the optimal reconstruction modes are selected by pattern recognition and smoothness to form the reconstructed signal; 3) the signal is identified by energy obtained by the optimal pattern.

2.1. CEEMDAN algorithm

CEEMDAN algorithm is a derivative algorithm of EMD algorithm. It has the advantages of good separation effect of mode spectrum, less times of shielding iteration and low computational cost, and is often used to deal with non-stationary and nonlinear signals. The algorithm can be described as the following steps[10]:

1) Add Gaussian white noise $w^i(n)$ to the signal $x(n)$ to form $x(n)+\varepsilon_0 w^i(n)$ and execute it for I times, and then decompose these new signal by EMD algorithm. The first mode can be written as:

$$IMF_1'\ (n)=\frac{1}{I}\sum_{i=1}^{I} IMF_1^i(n) \tag{1}$$

where IMF_1^i denotes the first EMD mode at the i-th execution.

2) Calculate the first residue as:

$$r_1(n)=x(n)-IMF_1'\ (n) \tag{2}$$

3) Decompose the realizations $r_1(n)+\varepsilon_1 E_1(w^i(n)),\ \ \mathrm{i}=1\cdots I$,where $E_i(\bullet)$ is a operator to produce the i-th mode by using EMD algorithm, until to get the first EMD mode and then the second mode can be defined as:

$$IMF_2'(n)=\frac{1}{I}\sum_{i=1}^{I}E_1(r_1(n)+\varepsilon_1E_1(w^i(n)))\tag{3}$$

4) For $k=2,\cdots,K$,calculate the k-th residue:

$$r_k(n)=r_{k-1}(n)-IMF_k'(n)\tag{4}$$

5) Continue to decompose the realizations $r_k(n)+\varepsilon_kE_k(w^i(n)),i=1\cdots I$,until to get their first EMD mode and the (k+1)-th mode as:

$$IMF_{k+1}'(n)=\frac{1}{I}\sum_{i=1}^{I}E_1(r_k(n)+\varepsilon_kE_k(w^i(n)))\tag{5}$$

6) Go to step (4) for next mode,until the residue can not be decomposed.

Repeat steps 4 to 6 until the remaining part no longer satisfies the decomposition condition (the remaining part does not contain at least two extrema), and finally the final residue is:

$$R(n)=x(n)-\sum_{k=1}^{K}IMF_k'\tag{6}$$

K is the number of decomposed modes, and the raw signal can be expressed as:

$$x(n)=\sum_{k=1}^{K}IMF_k'+R(n)\tag{7}$$

In the above formulas, ε_k is the standard deviation of noise added to the signal and it allows to select SNR in every stage. CEEMDAN decomposes the measured signal into multiple IMFs, which are generally from high frequency to low frequency. The magnetic dipole signal usually contains only several IMFs, so it is necessary to find out the IMFs containing signal components to reconstruct a new signal which is high signal to noise ratio (SNR).

2.2. Pattern recognition algorithm

Mode selection is an important part of CEEMDAN filtering algorithm, which determines the effect of filtering. The methods of mode selection that have been pointed out in some literature include: similarity and smoothness, correlation coefficient, wavelet soft and hard threshold, information entropy, Hausdorff distance and so on. Considering that the magnetic dipole target signal can generate magnetic target signal space by transforming the magnetic moment and geomagnetic field in the formula of magnetic dipole field, a signal pattern library containing a limited number of signal patterns can be obtained by using the similarity type. Therefore, the correlation between the signal patterns and the reconstructed signal can be used to select the best reconstruction modes. Moreover, once the signal pattern library is determined, it can be used all the time and hence it will not affect the calculation speed of the detection algorithm.

The typical detection scenario is: the platform carries a high sensitivity scalar magnetometer to move at a constant velocity v along a straight line in magnetic detection; a magnetic dipole target with magnetic moment $\mathbf{M}=M\hat{\mathbf{M}}$ is situated at a place where the distance to the moving track is R_0 . The magnetic anomaly signal measured by the sensor is

$$S(t)=\frac{\mu_0M}{4\pi}\left[\frac{3(\hat{\mathbf{T}}\cdot\mathbf{R})(\mathbf{R}\cdot\hat{\mathbf{M}})}{R^5}-\frac{(\hat{\mathbf{T}}\cdot\hat{\mathbf{M}})}{R^3}\right]\tag{8}$$

where $\hat{\mathbf{T}}$ is the direction of local geomagnetic field, and $\mathbf{R}$ is the real-time radial vector from the target to the sensor. The signal can be normalized to become $\overline{S}(t)$. Because the platform's velocity and heading direction, the local geomagnetic direction and the target orientation are variables, the normalized waveform $\overline{S}(t)$ can take any shape. To construct an optimal sample library, we first produce a massive sample library by varying each of the heading direction, geomagnetic direction and target orientation at a step of 2 Degrees. Then we reduce the number of samples based on similarity for a prescribed relevancy. We define the similarity index (SI) of two signals by

$$\rho(S_1, S_2) = \frac{Cov(S_1, S_2)}{\sqrt{Var(S_1)Var(S_2)}} \tag{9}$$

where *Cov* and *Var* denote the covariance and variance, respectively. For example, for SI of 0.8, there are only 11 samples, which are sufficient to represent any a magnetic anomaly signal waveform by 80% accuracy. For a SI of 0.9, the number of samples is 23, and for a SI of 0.99, the number of samples is 214, i.e., the 214 samples can simulate any a magnetic anomaly signal by 99% accuracy. The SI of 0.9 is chosen to construct a signal pattern library to detect target signal considering the calculation and accuracy. In Fig. 1, the 23 samples are shown, which means that any a real MDTS must be matching with one of these 23 samples at greater 90% similarity.

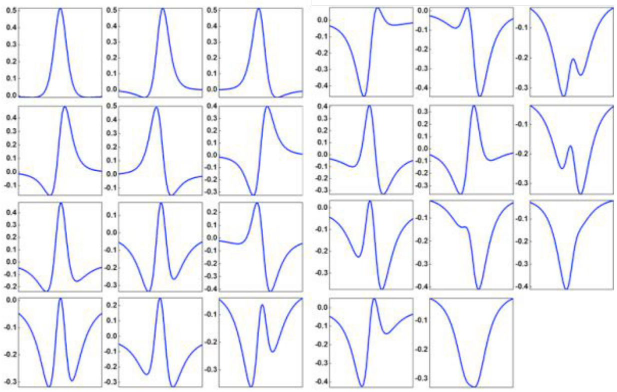


Fig. 1. The 23 signal samples for classification with similarity of 90%.

2.3. CEEMDAN-PR detection algorithm

In order to overcome the difficulty of detecting the MDTS under low signal noise ratio, this paper proposes a method combining CEEMDAN algorithm with pattern recognition algorithm. The CEEMDAN algorithm is used to decompose the measured signal, and the pattern recognition algorithm is used to select the useful modes for reconstruction. In addition, the signal pattern matching with the measured signal is determined in the process of selecting the number of reconstruction modes. The signal energy is obtained by the determined pattern and signal to determine whether the target exists or not.

The execution steps of this algorithm are listed as follows:

(1) Firstly, a high pass filter (cut-off frequency of 0.02Hz) is used to remove the low-frequency signal from the measured data;

(2) The filtered signal is decomposed into several intrinsic mode functions by CEEMDAN algorithm;

(3) Any number of modes are selected among the decomposed modes to reconstruct signal set;

(4) According to a new mode selection method (*FM*) combining signal similarity (*SSI*) calculated by the pattern recognition algorithm and the smoothness index (*ASMSE*) of reconstructed signal[11], the appropriate reconstructed signal can be selected out from the signal set. And with this method, the best matching pattern of the measurement signal can also be determined;

$$FM = \beta SSI + (1-\beta) ASMSE \tag{10}$$

Where β is the weight parameter which determines whether smoothness or similarity is more important in the final reconstructed signal and is usually set as 0.5.

(5) The energy is obtained by the interactions between the best matching pattern and the signal to detect, and then the threshold value is used to determine whether there exists a target.

3. Experiment

In the experiment, the target magnetic signal is generated by the formula (8). The earth's magnetic field is assumed to be (41644,7342.9,-15391) nT. The target is located at (0,0,0) and its magnetic moment is set as (43118,3772.3,43283) $\mathrm{A \cdot m^2}$. The measurement sensor is moving along x axis at the altitude of 600m to collect the magnetic data, the motion velocity of sensor is 100m/s, the sample ratio is 10Hz. In order to test the performance of the proposed method in this paper, Gaussian white noise and colored noise are respectively added to the simulation signal to form the synthetic signal to detect.

The synthetic signal with Gaussian white noise are shown in Fig.2. The signal to noise ratio (SNR) defined as formula (10) is -13.5dB in the synthetic signal and it is very obvious that the target signal has been completely submerged. In this experiment, $\varepsilon_k = 0.03$ in CEEMDAN algorithm.The processing results of EMD-PR algorithm and CEEMDAN-PR algorithm are shown in Fig.3-(a) and Fig.3-(b), respectively. In Fig.3-(a), the curves in the upper plot is the reconstructed signals obtained by combination of EMD-PR algorithm and different mode selection methods. Among the curves, the gray curve is obtained based on the criterion of continuous mean square error (CMSE)[12], the yellow curve is obtained by the criterion of similarity and smoothness(SSC)[11], the blued curve is obtained by the criterion proposed in this paper and the red curve is the target signal. Through comparison, it can be clearly recognized that the curve obtained by the new method is closer to that of pure target signal among the three curves. The bottom plot in Fig.3-(a) is the energy curve calculated by the optimal signal pattern. Fig.3-(b) is the processing result of CEEMDAN-PR algorithm.By comparing (a) and (b) of Fig.3, it can be seen that the reconstructed signal obtained by CEEMDAN-PR algorithm is smoother and more similar to the target signal, and the SNR of energy for signal detection is higher.

$$\mathrm{SNR} = 20\log_{10}\frac{S}{N}\mathrm{dB} \tag{11}$$

Where, S denotes the peak to peak value of signal and N denotes the peak to peak value of noise.

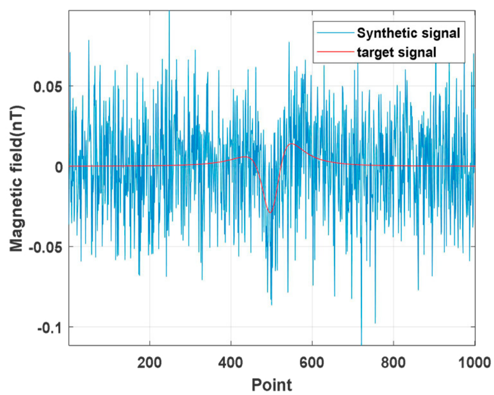


Fig. 2. The target signal and the synthetic signal with -13.5dB SNR.

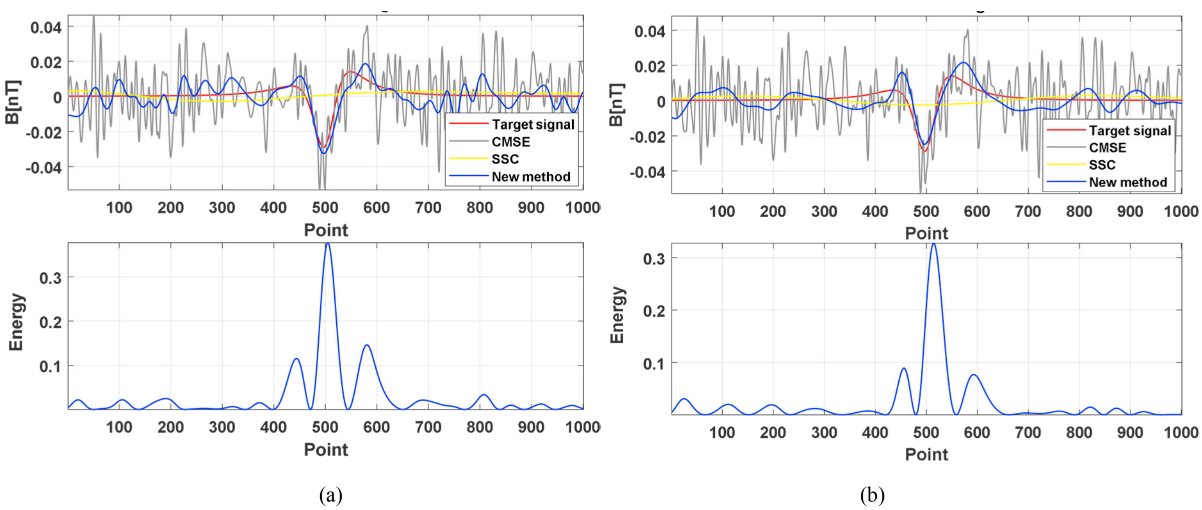


Fig. 3. The reconstructed signal and energy obtained by EMD-PR algorithm and CEEMDAN-PR algorithm.

Considering that the geomagnetic noise with the power spectral density of $1/f^{\alpha}$, $0<\alpha<2$ is the major magnetic noise source in the real magnetic target detection, the color magnetic noise with the power spectral density of $1/f^{\alpha}$, $\alpha=1.5$ is added to the target signal as the magnetic data to detect in the next experiment. The synthetic signal with SNR of -10dB in which the signal is submerged in colored noise is shown Fig.4. In this experiment, $\varepsilon_k=0.07$ in CEEMDAN algorithm. Fig.5 shows the processing results of two methods, EMD-PR algorithm and CEEMDAN-PR algorithm. It can be easily seen from the figures that the CEEMDAN-PR algorithm proposed in this paper is the most effective among the three methods to select the number of reconstruction modes and the reconstructed signal contains the most target signal information. It also can be seen from the energy diagram that SNR obtained by CEEMDAN-PR algorithm is higher than that of EMD-PR algorithm.

The consistency of the two experimental results shows that the detection algorithm, CEEMDAN-PR algorithm, can detect the MDTS well, and the effect of this method in processing the signal submerged by Gaussian white noise is better than that of the signal polluted by colored noise.

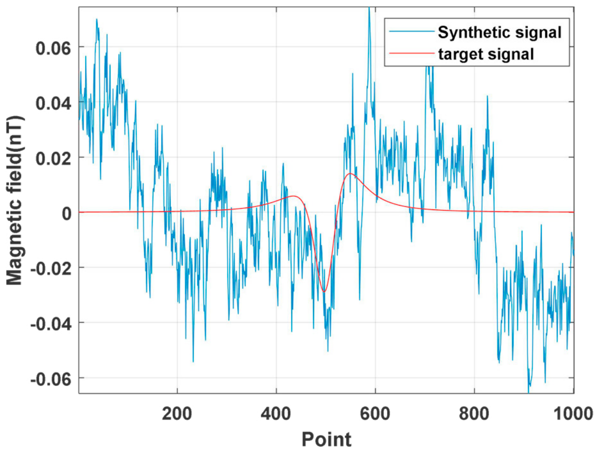
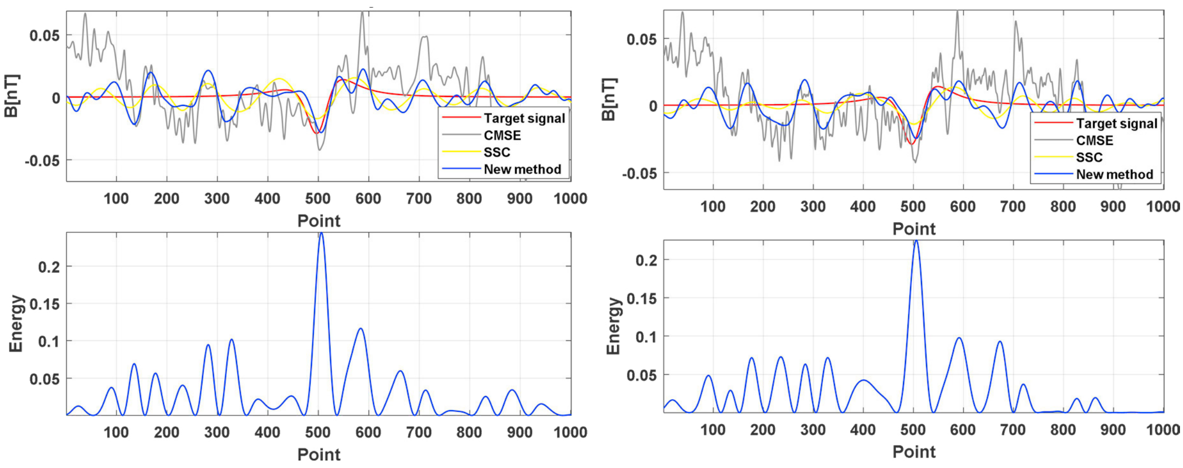


Fig. 4. The target signal and the synthetic signal with -10dB SNR.



(a) (b)

Fig. 5. The calculated results by EMD-PR algorithm and CEEMDAN-PR algorithm.

4. Summary and future work

In practice, the magnetic field generated by a long-distance magnetic target is usually submerged in the environmental magnetic noise. CEEMDAN algorithm has obvious advantages in processing non-stationary and nonlinear signal, and it overcomes the mode aliasing effect in EMD algorithm. Therefore, in this paper, it is used in the magnetic target signal detection to process the measured magnetic signal. How to select the appropriate number of modes to construct the best filtering signal is an important problem that has perplexed the application of CEEMDAN algorithm. In this paper, combined with the characteristics of magnetic target, a pattern recognition algorithm is used to select the optimal number of modes. The CEEMDAN-PR algorithm proposed in this paper can reduce the noise in the measured signal and improve the signal to noise ratio of detection data. Through the experimental comparison, it is proved that the algorithm is better than EMD-PR algorithm in the weak magnetic signal detection. In order to test the noise adaptability and stability of the algorithm, it will be used in the actual magnetic detection in the next work.

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